

# Foundation Program in Literacy, Numeracy and Skills

## Numeracy

### Grade 8

#### Lesson 14 and Lesson 15: Teacher's Guide

#### Foundation Program in Literacy, Numeracy and Skills

**A:** Understanding how to convert fractions to decimals to develop a deeper understanding of numbers and improve their ability to work with different mathematical forms in real life and academic situations

#### B. Competencies

Use numerical concepts in practical situations

**Element 4: Use fractions and decimals and percentages in authentic situations.**

**Performance Criteria:** Level 2: Convert decimal numbers into fractions and vice versa; solve simple problems involving decimals.

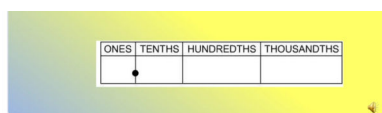
#### C. Learning Outcomes

At the end of the lesson, students should be able

- to develop a strong understanding of the relationship between fractions and decimals
- convert simple fractions to decimals using division
- convert simple terminating decimals to fractions using place value
- Students will be able to apply their knowledge of converting decimals and fractions to solve real- world problems

#### D. Resources and Materials:

- Place value



- Fraction and decimal flashcards
- Whiteboard/Markers

#### E. Implementation Guidelines

## NUMERACY

Teacher's Guide: GRADE 8- Lesson 14

Conversion – Decimal and Fractions

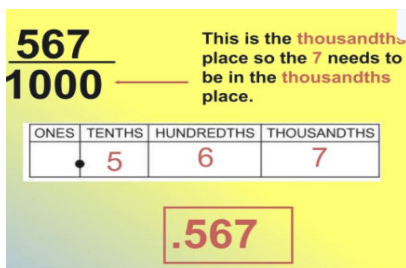
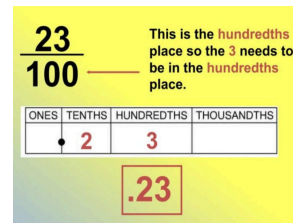
### How to convert Fractions to decimals?

#### Activity 1:

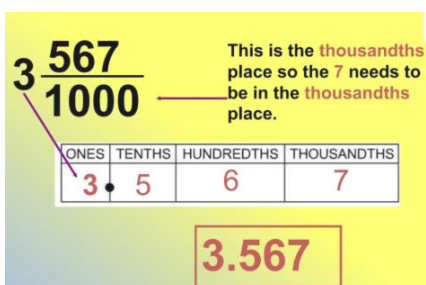
Conversion of fractions with denominators of 10, 100 and 1000 and so on using place value method.

Use the place value of the last digit in the number to determine what the denominator of the fraction will be.

$\frac{2}{10}$  → This is the tenths place so the 2 needs to be in the tenths place



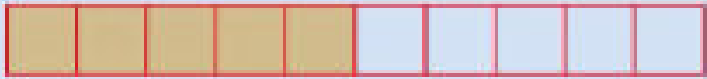
What if there is a whole number before the fraction?



Method 2: Another way to convert a fraction into decimal is to express it into an equivalent fraction with denominator either in 10, 100, 1000 etc.

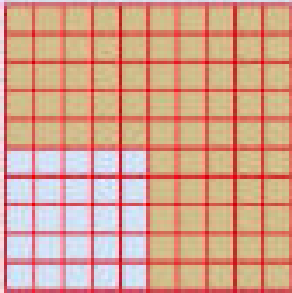
(a)  $\frac{1}{2}$  is equivalent to  $\frac{5}{10}$

$$\frac{1}{2} = \frac{5}{10}$$

$$\frac{5}{10} = 0.5$$


(b)  $\frac{3}{4}$  is equivalent to  $\frac{75}{100}$

$$\frac{3}{4} = \frac{75}{100} = 75 \times \frac{1}{100}$$

$$\frac{75}{100} = 0.75$$


Example:

Write (i)  $\frac{1}{2}$  (ii)  $\frac{3}{20}$  (iii)  $\frac{1}{8}$  as a decimal.

$$\begin{aligned} \text{(i)} \quad \frac{1}{2} &= \frac{1 \times 5}{2 \times 5} = \frac{5}{10} \\ &= \frac{5}{10} = 0.5 \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad \frac{3}{20} &= \frac{3 \times 5}{20 \times 5} = \frac{15}{100} \\ &= \frac{15}{100} = 0.15 \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad \frac{1}{8} &= \frac{1 \times 125}{8 \times 125} = \frac{125}{1000} \\ &= \frac{125}{1000} = 0.125 \end{aligned}$$

Teacher's Guide: GRADE 8- Lesson 15

Conversion – Decimal and Fractions

**How to convert Decimals to Fractions?**

**Activity 2:**

**To convert a decimal into fraction, we express it into fraction with denominator either in 10, 100, 1000 etc... .. .**

Example 1: Convert 0.3 to a fraction

Step 1: write the decimal as a fraction by dividing by 1

$$\frac{0.3}{1}$$

The number 0.3 is 3

We multiply the number 0.3 by 10 to get the integer value 3

Multiply both numerator and denominator by 10

$$0.3 = \frac{0.3 \times 10}{1 \times 10} = \frac{3}{10}$$

There is no common factor for 3 and 10, we cannot simplify

Example 2: Convert 0.43 as a fraction

Step 1: Write the decimal as a fraction divided by 1

$$\frac{0.43}{1}$$

The number 0.43 is the two hundredths

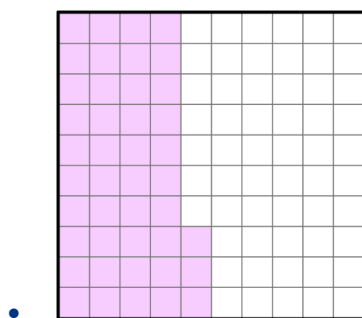
Multiply numerator and denominator by 100

$$\frac{0.43 \times 100}{1 \times 100} = \frac{43}{100}$$

No common factor for 43 and 100, we cannot further simplify

## Method 2:

The teacher can also use grid to represent 0.43 as a fraction



$$\frac{43}{100}$$

43 is represented by the number of shaded squares

## Method 3: The place value

### Decimals to Fractions

Converting **decimals to fractions** is when you represent a decimal as a fraction without changing its value.

 Example

Decimal  $\rightarrow$  0.43

| Ones | Tenths         | Hundredths      | Thousandths      |
|------|----------------|-----------------|------------------|
|      | $\frac{1}{10}$ | $\frac{1}{100}$ | $\frac{1}{1000}$ |
| 0    | 4              | 3               |                  |

$\times 10 \times \frac{1}{10}$

Read as **forty-three hundredths**.  
43 is the numerator of the fraction  
and 100 is the denominator  
because the last digit of the  
decimal is in the hundredths  
column of the place value chart.

$$0.43 = \frac{43}{100}$$

Decimal  $\rightarrow$  0.009

| Ones | Tenths         | Hundredths      | Thousandths      |
|------|----------------|-----------------|------------------|
|      | $\frac{1}{10}$ | $\frac{1}{100}$ | $\frac{1}{1000}$ |
| 0    | 0              | 0               | 9                |

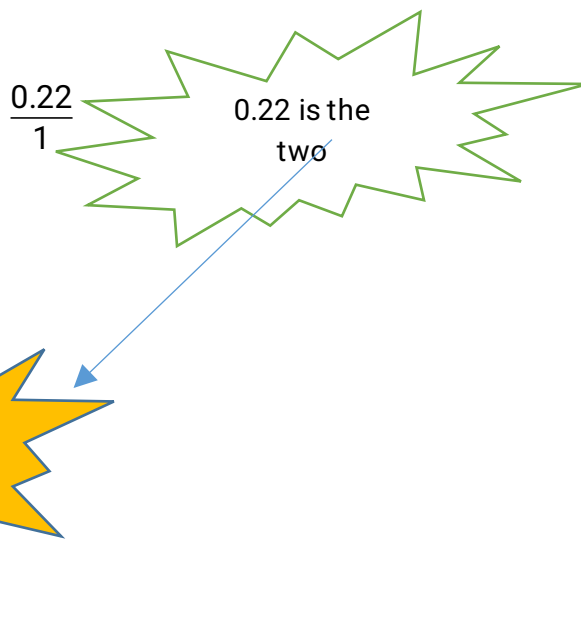
$\times 10 \times \frac{1}{10}$

Read as **nine thousandths**.  
9 is the numerator of the fraction  
and 1000 is the denominator  
because the 9 is in the thousandths  
column of the place value chart.

$$0.009 = \frac{9}{1000}$$



### Example 3: Convert 0.22 to a fraction



$$\frac{0.22 \times 100}{1 \times 100} = \frac{22}{100}$$

$$\frac{22 \div 2}{100 \div 2} = \frac{11}{50}$$

Divide both 22  
and 100 by a  
**common** factor 2

F. ASSESSMENT (in worksheet)

G. EXTENSION ACTIVITY